Working with Quantifiers: Translation, Laws and Search

This chapter is being written. *Motivating question:* "You know every quantifier rule. Can you turn a definition from analysis into a formula, negate it without error, and find the derivation it needs?" *S3-PLACEHOLDER: this chapter is being written; its index is assembled at S3.*

What you need. Chapters 1–5. *S3-PLACEHOLDER*

6.-1.1 Roadmap

- **Translating mathematics.** Definitions from analysis written as formulas with nested and bounded quantifiers, the bounded forms of the four rules, and the search method in one place.
- **Negating and moving quantifiers.** "Not every" is "some not", negation pushed inward quantifier by quantifier, and the laws that move a quantifier across a connective.
- **Uniformly continuous implies continuous, rule by rule.** The chapter's centerpiece: the full derivation, and the exact lines at which the converse breaks down.

6.-1.2 Named moves

- **Negate quantifier by quantifier:** push $\neg$ inward, turning $\forall$ into $\exists$ and back, and keep the bounds.
- **Read the dependency off the order:** a variable may depend on the variables quantified to its left, and only on them.